

Zhengyang Yu

Curriculum Vitae

San Diego & Bay Area

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA

Expected Graduation: June 2028

Relevant Coursework: Systems Programming, Computer Organization, Advanced Data Structures, Design and Analysis of Algorithms, Machine Learning

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*

- Built C++ firmware that authenticates multiple accounts, retrieves live data over HTTPS, and exposes reset windows, alerts, and interactive controls on an ESP32-S3 device.
- Automated rate-limit-aware polling with concurrent FreeRTOS tasks; added bounded structured logging, runtime diagnostics, and persistent configuration to improve reliability and troubleshooting.

2025 **HTML to ePub Converter**, *Personal Project*

- Developed a Python automation pipeline for HTML parsing, metadata generation, and ePub packaging with modular components and structured error handling.

2024 **Home Assistant Presence Plugin**, *Personal Project*

- Built a Python Home Assistant integration that converts entry and exit sensor events into real-time room-occupancy state.
- Automated smart-lighting workflows through REST APIs and explicit state transitions, improving reproducibility and simplifying defect diagnosis.

Skills

Programming Python, C++, C, Java

DevOps & Linux command line, Docker, Git, REST APIs, HTTPS, task automation, structured logging, Systems runtime diagnostics

Engineering Software Testing, Debugging, Configuration Management, Data Structures, Algorithms, Object-Oriented Design

Tools VS Code, PlatformIO

AI Tooling Codex-assisted implementation, refactoring, debugging, code validation, and technical documentation

Experience

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch

Diagnosed computer and software issues, documented solutions, and explained technical steps clearly to community members.

2024 Tutor, CTN (Community Technology Network)

Provided one-on-one tutoring to help adults improve digital literacy skills, including computer operation and internet usage.